Evaluation Performance:

We tested the model using Locust API to stress test the server with multiple concurrently active clients sending requests. We used it to see to find response time of the server for n number of concurrent clients and to see if the server has any exceptions or failures in the process.

For this purpose, we created a locustfile.py and implemented the code to use the gRPC server. Server was run on one terminal using “python server.py” and once the server had started, another terminal was opened and “locust” was run on that terminal to start the locust test.

We tested the server with 100 concurrent users where numbers of users increased from 1 to 100 in increments of 1 per second.

Here are the charts,

STATISTICS:

LOCUST

Host localhost:50051 | Status RUNNING | Users 100 | RPS 4.4 | Failures 0% | EDIT | STOP | RESET

STATISTICS | CHARTS | FAILURES | EXCEPTIONS | CURRENT RATIO | DOWNLOAD DATA | LOGS | LOCUST CLOUD

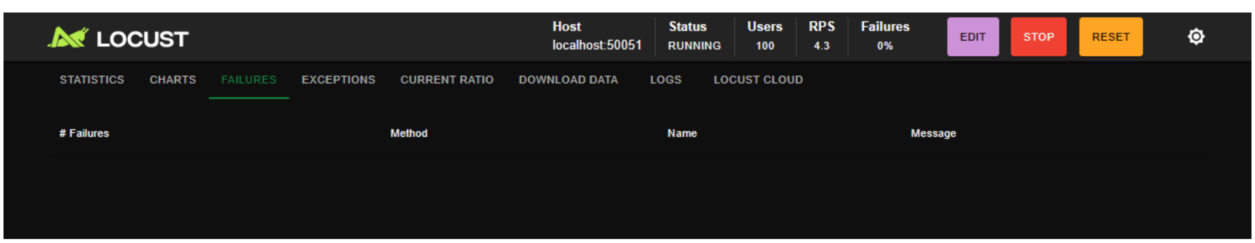
Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
grpc	GenerateSpeech	475	0	14000	20000	21000	12651.3	208	20929	112640	2.3	0
grpc	GenerateSpeechBase64	225	0	14000	20000	20000	12657.88	244	21044	112748	2.1	0
	Aggregated	700	0	14000	20000	21000	12653.42	208	21044	112674.71	4.4	0

We can see many statistics here like average response time in milliseconds for the server to respond to a request with various number of concurrent clients and Requests Per Second (RPS).

CHARTS:

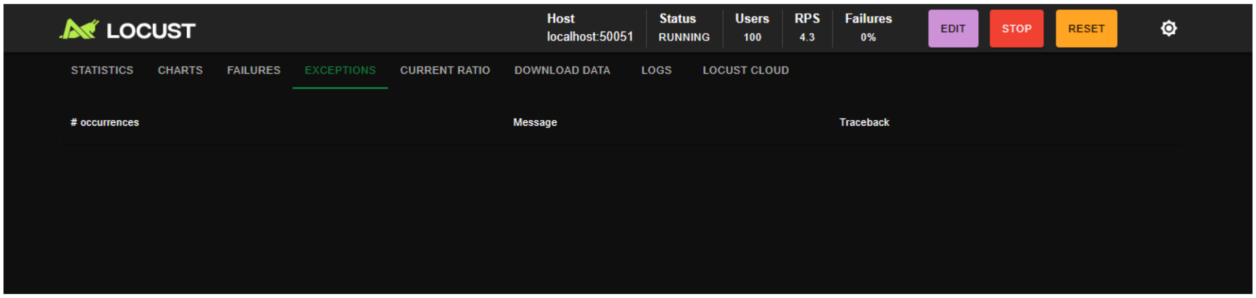
The chart shows many information like Total Requests per Second. Along with what is the RPS at each given time and number of failures at that time. As seen in the chart, there was no failures throughout the testing. We can also see that Response Time increased as more time passed because more users started to concurrently send requests with each passing second. The increase in number of Users as time passes is also shown in the chart above.

FAILURES:



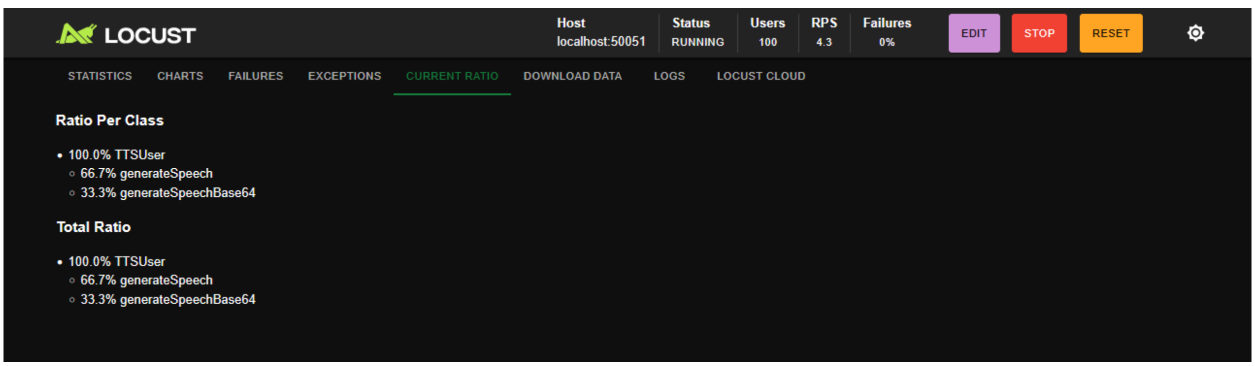
As you can see, there was no failure during the entire testing process.

EXCEPTIONS:



No exceptions were detected in the entire testing process.

RATIO:



My server has 2 methods, here is the ratio of requests made to each method during testing.

Other test cases can be found in the clients.py file.